

LC 990: Satisfiability of Equality Equations

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Target Companies: Amazon, UiPath

Examples & Dry Run

- `"a==b"`, `"b==a"` → **True**
- `"a==b"`, `"b!=a"` → **False**

`a==b` → `(a,b)` → merge

`b==c` → `(b,c)` → merge

`c!=a` → `(a!=c)` → **impossible**

"here we need just to know a, b, c are in some group"

Observation

process all `==` first, then check all the `!=`

DSU Initialization Logic

```
dsu() {  
    parent.resize(26);  
    size.resize(26,1);  
}
```

"as only 26 lower case characters are there, and there's nothing like index to pass to

`dsu(n)` "

Final Evaluation

After processing all `==` we now see `!=` if parent of both operands are same, it means they were merged before, hence **ret false**.

C++ Implementation

```

class dsu{
public:
    vector<int> parent;
    vector<int> size;
    dsu(){
        parent.resize(26);
        size.resize(26,1);

        for(int i=0;i<26;i++){
            parent[i]=i;
        }
    }

    int findparent(int node){
        if(parent[node]==node){
            return node;
        }

        parent[node]=findparent(parent[node]);
        return parent[node];
    }

    void unionit(int u,int v){
        int pu=findparent(u);
        int pv=findparent(v);

        if(pu==pv) return;

        if(size[pu]<size[pv]) swap(pu,pv);

        parent[pv]=pu;
        size[pu]+=size[pv];
    }
};

class Solution {
public:
    bool equationsPossible(vector<string>& equations) {

        dsu ds;
        for(auto &it:equations){
            if(it[1]=='='){
                ds.unionit(it[0]-'a',it[3]-'a');
            }
        }

        for(auto &it:equations){
            if(it[1]=='!'){
                if(ds.findparent(it[0]-'a')==ds.findparent(it[3]-'a')){
                    return false;
                }
            }
        }
    }
};

```

```
        }  
    }  
    return true;  
}  
};
```